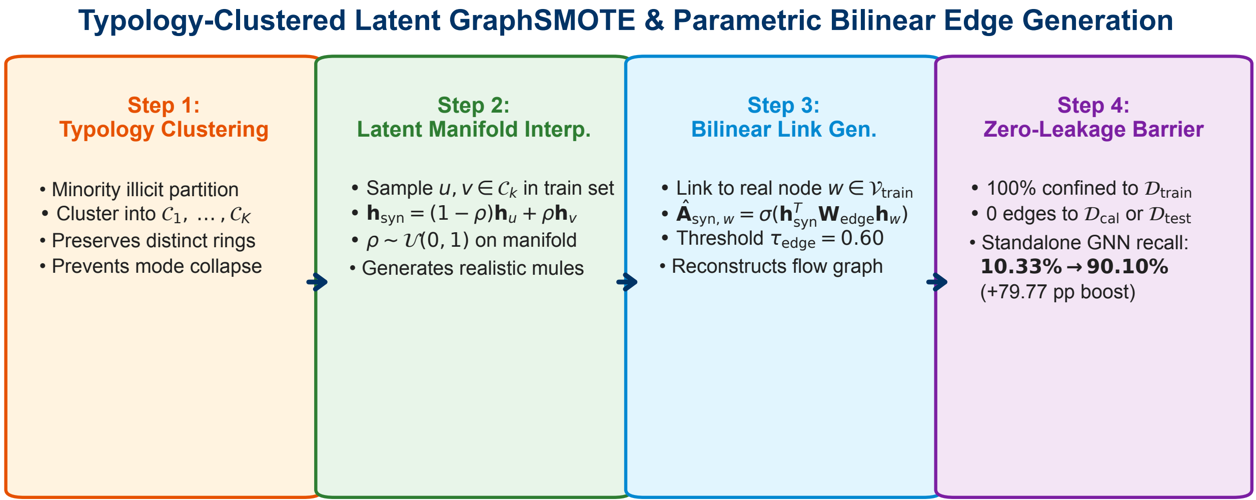
Typology-Clustered Latent GraphSMOTE & Parametric Bilinear Edge Generation
Step 1:
Typology Clustering
• Minority illicit partition
• Cluster into $\mathcal{C}_1, \ldots, \mathcal{C}_K$
• Preserves distinct rings
• Prevents mode collapse
Step 2:
Latent Manifold Interp.
• Sample $u, v \in \mathcal{C}_k$ in train set
• $\mathbf{h}_{\text{syn}} = (1-\rho)\mathbf{h}_u + \rho\mathbf{h}_v$
• $\rho \sim \mathcal{U}(0,1)$ on manifold
• Generates realistic mules
Step 3:
Bilinear Link Gen.
• Link to real node $w \in \mathcal{V}_{\text{train}}$
• $\hat{\mathbf{A}}_{\text{syn},w} = \sigma(\mathbf{h}_{\text{syn}}^T \mathbf{W}_{\text{edge}} \mathbf{h}_w)$
• Threshold $\tau_{\text{edge}} = 0.60$
• Reconstructs flow graph
Step 4:
Zero-Leakage Barrier
• 100% confined to $\mathcal{D}_{\text{train}}$
• 0 edges to $\mathcal{D}_{\text{cal}}$ or $\mathcal{D}_{\text{test}}$
• Standalone GNN recall:
10.33% → 90.10%
(+79.77 pp boost)